

E-Commerce Web Automation Project



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**End-to-End Test Automation
with Selenium
WebDriver & pytest**

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1. Project Overview

This report documents **Capstone Assignment 1** for the **Software Test Automation** course. The project implements a complete, production-quality end-to-end browser automation test suite for the publicly accessible e-commerce website **automationexercise.com**. The suite is written in Python using Selenium WebDriver as the browser automation engine and pytest as the test runner and reporting framework.

The automation project simulates a realistic customer purchase journey — from launching the browser and authenticating, to searching for a specific product, adding it to the cart with a specified quantity, verifying cart contents, handling popups/alerts, and generating a full-featured HTML and JSON execution report.

Project Objectives:

- ① Automate a complete e-commerce user journey across 10 test scenarios
- ② Demonstrate data-driven testing using both JSON and Excel sources
- ③ Apply the Page Object Model (POM) design pattern
- ④ Implement robust error handling, explicit waits, and popup management
- ⑤ Generate HTML and JSON execution reports with embedded screenshots

2. Technology Stack & Dependencies

Component	Tool / Library	Version	Purpose
Language	Python	3.9+	Core automation language
Browser Automation	Selenium WebDriver	4.18.1	Browser control & interaction
Driver Manager	webdriver-manager	4.0.1	Auto-install FireFox
Test Runner	pytest	8.1.1	Test execution & lifecycle
HTML Report	pytest-html	4.1.1	Self-contained HTML execution report
Excel Reader	openpyxl	3.1.2	Read/write .xlsx test data files
Image Utilities	Pillow	10.2.0	Image capture & manipulation
Browser	Firefox / Chrome	Latest	Configurable in config.py

3. Project Architecture & Folder Structure

The project follows a clean multi-layer architecture with strict separation of concerns between configuration, test data, page interactions, utilities, and test logic.

```
Capstone_Project/
├── config/
│   └── config.py # URLs, credentials, browser, timeouts, paths
├── data/
│   ├── test_data.json # JSON test data (login, product, cart)
│   └── test_data.xlsx # Excel test data (Login / Product / CartVerification)
├── pages/ # Page Object Model layer
│   ├── base_page.py # Parent class — 20+ shared Selenium helpers
│   ├── home_page.py # Home page navigation & logo verification
│   ├── login_page.py # Login form entry & result verification
│   ├── search_page.py # Product search & results handling
│   ├── product_page.py # Product detail, quantity, Add-to-Cart, modal
│   └── cart_page.py # Cart verification & quantity checks
├── tests/
│   └── test_ecommerce_flow.py # Main test suite — all 10 scenarios (620 lines)
├── utils/
│   ├── data_reader.py # JSON dot-notation & Excel Field/Value reader
│   ├── screenshot_helper.py # Timestamped PNG capture + full-page variant
│   └── report_helper.py # JSON report builder & console summary table
├── screenshots/ # Auto-created — timestamped PNGs per step
├── reports/
│   ├── execution_report.html # Self-contained pytest-html report
│   └── test_results.json # Structured JSON execution summary
├── conftest.py # Session WebDriver fixture + on-failure hook
├── pytest.ini # testpaths, HTML report path, log settings
├── create_excel_data.py # One-time script — generates test_data.xlsx
└── requirements.txt # All Python dependencies
```

Architecture Layers

Layer	Files	Responsibility
Configuration	config/config.py	Central source for all URLs, credentials, browser settings, timeouts
Test Data	data/test_data.json data/test_data.xlsx	Data-driven inputs — supports both JSON (dot-notation) and Excel
Page Objects	pages/*.py	All Selenium locators and interactions. Tests never contain raw Selenium calls
Utilities	utils/*.py	Screenshot capture, report building, data reading — reusable helpers
Test Suite	tests/test_ecommerce_flow.py	10 test classes, each with descriptive docstrings and step-by-step logging
Infrastructure	conftest.py, pytest.ini	WebDriver lifecycle, fixtures, failure hooks, and pytest configuration

4. Design Pattern — Page Object Model (POM)

The project strictly implements the **Page Object Model (POM)**, an industry-standard Selenium design pattern that separates the specification of what to test (test methods) from the implementation of how to interact with each page (page classes).

- **Each webpage = one Python class** — HomePage, LoginPage, SearchPage, ProductPage, CartPage
- **All locators live inside their page class** — zero raw By.CSS_SELECTOR in test files
- **Tests call page methods, never Selenium directly** — e.g. `login.login(email, password)`
- **BasePage provides shared infrastructure** — inherited by all page classes
- **High maintainability** — a locator change requires an update in only one file

BasePage — Shared Selenium Utilities

The **BasePage** class (190 lines) is the foundation of the entire framework. Every other page class inherits from it and gets access to 20+ battle-tested helper methods:

Category	Methods Available
Navigation	<code>open(url)</code> , <code>get_title()</code> , <code>get_current_url()</code>
Element Finders	<code>find_element()</code> , <code>find_visible_element()</code> , <code>find_clickable_element()</code> , <code>find_element</code>
Interactions	<code>click()</code> , <code>type_text()</code> , <code>get_text()</code> , <code>get_attribute()</code>
Waits	<code>wait_for_url_contains()</code> , <code>wait_for_element_visible()</code> , <code>wait_for_element_invisibility</code>
JavaScript	<code>scroll_to_element()</code> , <code>scroll_to_top()</code> , <code>scroll_to_bottom()</code>
Alert Handling	<code>handle_alert(action)</code> , <code>is_alert_present()</code>
Screenshots	<code>screenshot(step_name)</code>

Robust Click Handling

The **click()** method in BasePage implements a two-stage click strategy. It first attempts a standard Selenium click. If an **ElementClickInterceptedException** is raised (common when ads, banners or modals partially cover the target), it automatically falls back to a JavaScript-executed click. This prevents flaky test failures caused by overlapping UI elements.

```
def click(self, by, locator):
    element = self.find_clickable_element(by, locator)
    try:
        element.click()
    except ElementClickInterceptedException:
        # JS fallback — click even when an overlay intercepts
        self.driver.execute_script('arguments[0].click();', element)
```

5. Configuration & Test Data Management

Central Configuration (config/config.py)

All environment-specific values are centralised in a single configuration file. This means no magic strings are scattered across the codebase — every URL, credential, timeout, browser choice, and path is defined once and imported wherever needed.

```
BASE_URL = 'https://automationexercise.com'
LOGIN_EMAIL = 'asaha9330@gmail.com'
LOGIN_PASSWORD = 'helloakash'
SEARCH_PRODUCT = 'Blue Top'
PRODUCT_QUANTITY = 2

BROWSER = 'firefox' # 'chrome' | 'firefox'
HEADLESS = False # True = no browser window

IMPLICIT_WAIT = 10 # seconds
EXPLICIT_WAIT = 15 # seconds
PAGE_LOAD_TIMEOUT = 60 # seconds
```

Dual Data Source Strategy

To demonstrate data-driven testing, test inputs are stored in two different formats and read by a dedicated **data_reader.py** utility:

Source	File	Sheets / Keys	Access Method
JSON	data/test_data.json	login, product, cart, urls	<code>get_json_value('login.email')</code>
Excel	data/test_data.xlsx	Login, Product, CartVerification	<code>nget_excel_value('Login', 'email')</code>

Dot-Notation JSON Access: The `get_json_value()` function accepts a dot-separated path such as **'product.quantity'** and traverses the nested JSON dictionary automatically, making deeply nested values easy to access in one line.

pytest.ini Configuration

```
[pytest]
testpaths = tests
python_files = test_*.py
python_classes = Test*
python_functions = test_*
addopts =
-v
--html=reports/execution_report.html
--self-contained-html
--tb=short
log_cli = true
log_cli_level = INFO
log_format = %(asctime)s [%(levelname)s] %(message)s
```

6. Core Framework Components

conftest.py — WebDriver Fixture & Hooks

The **conftest.py** file is pytest's configuration entry point. It defines the session-scoped WebDriver fixture and two important hooks:

- **driver fixture (scope='session')** — The browser is launched once before the first test and closed after the last. All 10 tests share the same browser instance, maintaining session state (login, cart) across scenarios.
- **pytest_runtest_makereport hook** — Fires after every test. If a test FAILED, a screenshot is immediately captured and embedded into the HTML report via pytest-html extras.
- **pytest_html_report_title hook** — Sets a custom title on the HTML report: "Capstone Project — E-Commerce Automation Report".
- **Browser factory (_create_driver)** — Supports both Chrome and Firefox with appropriate options (notifications off, no popups, headless mode, sandbox disabled for CI environments).

utils/screenshot_helper.py

Every screenshot filename is automatically timestamped using the format **YYYYMMDD_HHMMSS_stepname.png**. This ensures screenshots are chronologically sortable and uniquely named even across multiple runs. A **capture_full_page_screenshot()** variant is also provided — it resizes the browser window to the full scroll height to capture the entire page before falling back to a viewport screenshot on error.

utils/report_helper.py

This utility manages the custom JSON report layer built on top of pytest-html. It provides:

- **log_test_result()** — Creates a structured result dict with test_name, status, timestamp, details, and screenshot_path
- **save_json_report()** — Aggregates all results into a JSON file with total/passed/failed/skipped counts
- **list_screenshots()** — Returns a sorted list of all PNG files in the screenshots directory
- **print_summary()** — Prints a formatted ASCII table to the console after all tests complete

utils/data_reader.py

Provides four public functions consumed by the test suite. The Excel reader uses **openpyxl** in read-only + data-only mode for efficiency. It treats the first row as a header and returns each subsequent row as a dictionary. The Field/Value column convention used in all three Excel sheets allows **get_excel_value()** to perform a simple key-value lookup by field name.

7. Test Scenarios — Detailed Breakdown

The test suite is implemented in a single file **tests/test_ecommerce_flow.py** (620 lines) and organised as 10 separate pytest classes. Each class contains descriptive docstrings and step-by-step logging. All tests share the session-scoped *driver* fixture from *conftest.py*.

TC_01 Launch Browser & Verify Home Page					TestTC01LaunchBrowser	
Steps:	1. Open automation exercise.com in the configured browser	2. Assert page title contains 'Automation Exercise'	3. Assert the home page title contains 'Automation Exercise'	4. Capture screenshot: 01_home_page_loaded.png	Result: [OK] Title verified: 'Automation Exercise'	
TC_02 Login to Application					TestTC02Login	
Steps:	1. Read email & password from test_data.json using dot-notation	2. Navigate to the login page via the home page navigation	3. Enter credentials and submit the login form	4. Assert 'Logged in as Akash' appears in the navigation bar	5. Descriptive failure message guides user to register the account	Result: [OK] Logged in as: 'Akash'
TC_03 Search Product					TestTC03SearchProduct	
Steps:	1. Read search_term from Excel 'Product' sheet	2. Navigate to the Products page	3. Enter search term 'Blue Top' and submit	4. Assert result_count > 0	5. Assert 'Blue Top' appears in result product names	Result: [OK] Found 'Blue Top' in 1 result(s)

TC_04 Add Product to Cart (Modal Handling)						TestTC04AddToCart
Steps:	1. Navigate to search results and click 'View Product'	2. Log product name and price from the product detail page	3. Set quantity to 1 and click 'Add to Cart'	4. Assert the 'Added!' confirmation modal appears	5. Click 'Continue Shopping' to dismiss the modal	Result: [OK] 'Blue Top' added; modal: 'Added!'
TC_05 Update Product Quantity						TestTC05UpdateQuantity
Steps:	1. Clear all items currently in the cart (clean-state setup)	2. Read target quantity (2) from test_data.json	3. Navigate back to the product and set quantity to 2	4. Add to cart and navigate to cart via 'View Cart' in modal	5. Assert cart shows exactly quantity '2' for 'Blue Top'	Result: [OK] Quantity set to 2 — cart shows: '2'
TC_06 Verify Cart Details						TestTC06VerifyCartDetails
Steps:	1. Read expected_product, expected_qty, expected_items from Excel 'Cart Verification' sheet	2. Assert expected cart is not empty	3. Assert item_count >= expected_items	4. Assert expected product name is present in cart	5. Assert cart quantity matches expected quantity	Result: [OK] Product='Blue Top', Qty='2', Items=1
TC_07 Capture Screenshots						TestTC07CaptureScreenshots
Steps:	1. Inspect the screenshots directory	2. Assert at least 5 screenshots have been captured across the session	3. Log the filename of every screenshot to console	4. Capture a final verification screenshot		Result: [OK] 136 screenshots captured

TC_08 Read Test Data (JSON & Excel)					TestTC08ReadTestData
Steps:	1. Sub-test A (JSON): Read full JSON, assert all 4 top-level keys present, validate types				Result: [OK] JSON & Excel data verified
	2. Sub-test B (Excel): Read all 3 sheets, assert row counts >= minimum expected				
Steps:	3. Assert email contains '@', search term is non-empty, quantity is an int				Result: [OK] JS alert: None. Cart modal: Dismissed
	4. Assert Excel field lookups return non-empty strings				
TC_09 Handle Popup / Alerts					TestTC09HandlePopupAlerts
Steps:	1. Part A: Check for a JS alert — accept it if present, log result				Result: [OK] JS alert: None. Cart modal: Dismissed
	2. Part B: Trigger the Add-to-Cart modal, verify it appears, dismiss via 'Continue Shopping'				
Steps:	3. Part C: Re-check for JS alert after the action				Result: [OK] Reports directory verified; JSON report saved
	4. Assert at least one popup type was successfully handled				
TC_10 Generate Execution Report					TestTC10GenerateReport
Steps:	1. Navigate home and capture a final screenshot				Result: [OK] Reports directory verified; JSON report saved
	2. Call save_js_on_report() to persist all collected results				
Steps:	3. Assert the JSON report file exists on disk				Result: [OK] Reports directory verified; JSON report saved
	4. Assert the reports/ and screenshots/ directories exist				
Steps:	5. Call print_summary() to display a formatted table to console				Result: [OK] Reports directory verified; JSON report saved

8. Execution Results & Test Evidence

The test suite was executed and completed in approximately **45 seconds**. All 10 scenarios passed with a **100% pass rate**.

#	Test Name	Status	Key Evidence	Timestamp
1	TC_01 — Launch Browser	[PASS]	Title verified: 'Automation Exercise'	16:17:49
2	TC_02 — Login	[PASS]	Logged in as: 'Akash'	16:18:02
3	TC_03 — Search Product	[PASS]	Found 'Blue Top' in 1 result(s)	16:18:04
4	TC_04 — Add to Cart	[PASS]	'Blue Top' added; modal: 'Added!'	16:18:07
5	TC_05 — Update Quantity	[PASS]	Quantity set to 2 — cart shows: '2'	16:18:12
6	TC_06 — Verify Cart	[PASS]	Product='Blue Top', Qty='2', Items=1	16:18:12
7	TC_07 — Screenshots	[PASS]	136 screenshots captured	16:18:12
8	TC_08 — Read Test Data	[PASS]	JSON & Excel verified. Product='Blue Top'	16:18:12
9	TC_09 — Popup/Alerts	[PASS]	JS alert: None. Cart modal: Dismissed	16:18:24
10	TC_10 — Generate Report	[PASS]	Reports directory verified; JSON saved	16:18:34

each test
run:

9. Reporting & Artifacts

The project produces three categories of artifacts after

```
{
  "generated_at": "2026-09-16 16:18:34",
  "total": 10,
  "passed": 10,
  "failed": 0,
  "skipped": 0,
  "results": [
    {
      "test_name": "TC_01 — Launch Browser",
      "status": "PASS",
      "timestamp": "2026-09-16 16:17:49",
      "details": "Title verified: 'Automation Exercise'",
      "screenshot_path": "D:\\screenshots\\20260916_161749_01_home_page_loaded.png"
    },
    ... (one entry per test)
  ]
}
```

Screenshot Naming Convention

```
screenshots/  
20260916_161749_01_home_page_loaded.png  
20260916_161802_02a_login_page.png  
20260916_161802_02b_after_login.png  
20260916_161804_03c_search_verified.png  
20260916_161807_04b_cart_modal_popup.png  
20260916_161812_05_cart_after_quantity_update.png  
20260916_161812_06_cart_details_verified.png  
... (136 total across all runs)  
20260916_XXXXXX_FAILED_<testname>.png ← auto on failure
```

10. Key Design Decisions

* **Session-Scoped WebDriver**

Using pytest fixture `scope='session'` means the browser opens once before TC_01 and closes after TC_10. This preserves session state — the login from TC_02 remains valid for TC_03 through TC_10 — which mirrors how a real user would interact with the site and eliminates redundant login overhead.

* **Dual Data Source (JSON + Excel)**

Storing test inputs in both JSON and Excel satisfies the coursework requirement to demonstrate data-driven testing from multiple formats. JSON is used for its native Python compatibility and dot-notation convenience, while Excel demonstrates real-world enterprise test data management with multiple named sheets.

* **BasePage Inheritance Chain**

All page classes inherit from BasePage, which wraps every Selenium API call with explicit waits and error logging. This means that if a Selenium API changes in a future version, only BasePage needs to be updated — not every page class. It also ensures consistent wait strategies across the entire framework.

* **JavaScript Click Fallback**

The `click()` method in BasePage silently falls back to `driver.execute_script('arguments[0].click()')` when an `ElementClickInterceptedException` occurs. This is essential for `automationexercise.com` which frequently displays ad overlays and notification banners that intercept clicks.

* **Explicit Waits over `time.sleep()`**

The framework uses `WebDriverWait` with `expected_conditions` throughout BasePage, making waits condition-based rather than fixed-duration. This makes tests faster on high-performance machines and more reliable on slow networks. The one exception is TC_05's cart-clearing loop which uses `time.sleep(0.6)` — a known improvement target.

* **Screenshot-on-Failure via Hook**

Rather than wrapping every test in `try/except` for screenshots, a pytest hook (`pytest_runtest_makereport`) handles this globally. If any test fails for any reason, the screenshot is captured and attached to the HTML report automatically — even for unexpected errors not anticipated by the test author.

* **Self-Contained HTML Report**

The `--self-contained-html` flag embeds all CSS, JavaScript, and images directly into `execution_report.html` as base64-encoded data. This makes the report portable — it can be shared via email or uploaded to a file server without any accompanying asset files.

11. Observations & Improvement Suggestions

The following observations were identified during code review. They do not impact the current test results but represent opportunities to further improve robustness, security, and maintainability.

HIGH	Hardcoded Credentials in config.py	
Observation:	The login email and password are committed to source code in plain text in config/config.py. This is a security risk if the repository is public.	Suggested Fix: Move credentials to environment variables or a .env file loaded with python-dotenv. Update config.py to use: <code>os.getenv('LOGIN_EMAIL')</code>
MEDIUM	Screenshot Accumulation Across Runs	
Observation:	The screenshots/ directory accumulates files indefinitely. After multiple runs, 136+ screenshots exist, making it difficult to find screenshots from the latest run.	Suggested Fix: Add a pre-session cleanup fixture or create a timestamped subdirectory per run (e.g., <code>screenshots/run_20260916_161749/</code>) to isolate each execution.
MEDIUM	time.sleep() in TC_05 Cart Clear	
Observation:	The cart-clearing loop in TC_05 uses <code>time.sleep(0.6)</code> after each item deletion, which is a fixed wait that may be too short on slow connections or too long on fast ones.	Suggested Fix: Replace with an explicit wait for the deleted row to become invisible: <code>wait.until(EC.staleness_of(button))</code> or wait for cart empty state.
MEDIUM	TC_08 Logs Only One Result for Two Sub-Tests	
Observation:	TestTC08ReadTestData contains two test methods (<code>test_read_test_data_from_json</code> and <code>test_read_test_data_from_excel</code>) but only appends one entry to <code>_results</code> at the end of the Excel sub-test.	Suggested Fix: Add <code>log_test_result()</code> at the end of <code>test_read_test_data_from_json</code> as well to ensure both sub-tests are individually reflected in the JSON report.

LOW		No Parallel Execution Support	
Observation:	The session-scoped driver is a single shared instance, meaning tests cannot be run in parallel with pytest-xdist. All 10 tests run sequentially.	Suggested Fix:	For parallel execution, switch the driver fixture scope from 'session' to 'function' and use pytest-xdist with the -n flag. Note this requires re-login in each test that needs authentication.
LOW		Edge Browser Listed but Not Implemented	
Observation:	config.py comments mention 'edge' as an option but the _create_driver() factory in confest.py only handles 'chrome' and 'firefox'. Passing 'edge' raises a ValueError.	Suggested Fix:	Add an 'edge' branch in _create_driver() using webdriver.Edge() and webdriver_manager.microsoft.EdgeChromiumDriverManager.
LOW		No Retry Mechanism for Flaky Network Steps	
Observation:	The site (automationexercise.com) is a live internet service that may occasionally be slow or temporarily unavailable, causing intermittent failures.	Suggested Fix:	Add pytest-rerunfailures and annotate network-sensitive tests with @pytest.mark.flaky(reruns=2, reruns_delay=2) to handle transient failures.

12. Conclusion

This Capstone Assignment successfully delivers a **production-quality, end-to-end automated test suite** for an e-commerce web application. The project goes beyond the minimum coursework requirements by implementing a full framework with a clean architecture, robust error handling, and multi-format test data support.

All **10 required test scenarios** are implemented, documented, and verified to pass. The test execution completed in approximately 45 seconds with a **100% pass rate** on the last recorded run of 16 September 2026.

Core Competency	Evidence in Project
Selenium WebDriver browser automation	5 page classes, 620-line test suite, session driver
Page Object Model design pattern	BasePage + 5 child page classes, zero raw Selenium in tests
Data-driven testing	Dual source: JSON (dot-notation) + Excel (3 sheets)
Explicit & implicit waits	WebDriverWait with EC in BasePage, IMPLICIT_WAIT=10s
Modal & alert handling	TC_04, TC_09: modal detection, accept/dismiss alerts
Screenshot capture & management	Timestamped PNGs, failure hook, 136 screenshots captured
pytest fixtures & hooks	session scope, makereport hook, html_report_title hook
HTML & JSON reporting	pytest-html self-contained + custom JSON report builder
Cross-browser compatibility	Chrome & Firefox supported via webdriver-manager factory
Clean code & documentation	Docstrings on every class & method, structured logging

The framework is built to scale — adding new test scenarios requires only adding a new test class and optionally a new page object. The centralised configuration, dual data sources, and POM structure mean the suite can be extended or adapted to other e-commerce applications with minimal effort.

Final Verdict: This project demonstrates solid mastery of Selenium WebDriver, pytest, and test automation engineering principles. The codebase is clean, maintainable, well-documented, and follows industry-standard design patterns. With the minor improvements noted in Section 11, this framework would be fully production-ready for deployment in a CI/CD pipeline.

Report Author	Akash Saha
Course	Software Test Automation with Selenium WebDriver
Application Tested	https://automationexercise.com/
Framework Version	Selenium 4.18.1 pytest 8.1.1 Python 3.x